PATEL MART B.V.

DemandIQ NL

Project Blueprint & Architecture Document

Data Dictionary · KPI Definitions · Page Wireframes · Schema Design · View Logic · Risk Score Architecture · Naming Conventions · Glossary

Document Type	Project Blueprint & Technical Reference
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1. Document Purpose & How to Use This

This document is the single authoritative technical reference for the DemandIQ NL system built at Patel Mart B.V. It is not a project report (see the Analytics Report for findings and recommendations). It is an architect document — the reference you reach for when building, extending, debugging, or handing over any component of the system.

Intended audiences and primary use cases:

Reader	What to refer to
Data Analyst (current/future)	Data Dictionary, View Definitions, Risk Score Logic, Naming Conventions
Data Engineer	Pipeline Data Flow, Schema Diagram, View Definitions, Naming Conventions
BI Developer / Power BI Developer	KPI Definitions, Page Wireframes, View Definitions
Supply Chain Manager	KPI Definitions, Glossary, Page Wireframes
New team member onboarding	Full document — read in section order

> **Versioning note:** This document should be updated whenever the schema, views, KPI definitions, or dashboard layout changes. Version history should be maintained in the project GitHub repository alongside the code.

2. Pipeline Data Flow

The DemandIQ NL pipeline moves data through five distinct layers. Each layer has a defined input, transformation, and output. No layer writes backwards into a prior layer.

#	Layer	Input	Output	Tool
1	**Source**	M5 Kaggle CSVs (3 files)	Raw files on disk	File system
2	**Staging**	Raw CSVs via chunked stream	58.3M rows across 3 staging tables	Python + psycopg2
3	**Warehouse**	Staging tables via SQL transformation	37.2M fact rows + 4 dim tables	SQL + Python
4	**Engine**	Warehouse tables via Jupyter notebook	fact_inventory: 23,311 rows with risk scores	Python + psycopg2
5	**Analytical Views**	Warehouse + fact_inventory via SQL	5 PostgreSQL views for Power BI	SQL
6	**Dashboard**	5 PostgreSQL views via Npgsql	5-page Power BI executive dashboard	Power BI + Npgsql

Data flow direction (read-only across layers):

```
CSV Files → stg_* tables → dim_* + fact_sales → fact_inventory →
vw_* views → Power BI
```

> **Rule:** The staging layer is append-only. The warehouse layer is rebuilt from staging on each full refresh. fact_inventory is rebuilt from warehouse on each engine run. Views are always live queries against the underlying tables.

3. Schema Diagram & Relationships

3.1 Staging Schema (schema: staging)

Table	Rows (approx.)	Description
stg_sales	**58,327,370**	One row per item × store × date. Wide CSV melted to long format.
stg_prices	**6,841,121**	Weekly sell price per item × store.
stg_calendar	**1,969**	One row per date. Dutch public holiday flags applied.

3.2 Warehouse Schema (schema: warehouse)

Table	Type	Rows	Primary Key
fact_sales	**Fact**	37,200,000+	date_sk + product_sk + store_sk (composite)
fact_inventory	**Fact**	23,311	sku_id + store_sk (composite)
dim_date	**Dimension**	1,969	date_sk (integer surrogate)
dim_product	**Dimension**	3,049	product_sk (integer surrogate)
dim_store	**Dimension**	10	store_sk (integer surrogate)
dim_price	**Dimension**	6,841,121	item_id + wm_yr_wk (composite)

3.3 Join Relationships & Cardinality

Left Table.Column	→	Right Table.Column	Cardinality
fact_sales.date_sk	→	dim_date.date_sk	Many-to-One
fact_sales.product_sk	→	dim_product.product_sk	Many-to-One
fact_sales.store_sk	→	dim_store.store_sk	Many-to-One
fact_sales.product_sk (via dim_product.item_id + wm_yr_wk)	→	dim_price.item_id + wm_yr_wk	Many-to-One
fact_inventory.sku_id	→	dim_product.product_sk	Many-to-One
fact_inventory.store_sk	→	dim_store.store_sk	Many-to-One
dim_product.item_id	→	stg_prices.item_id	One-to-Many (bridge)

> **Important join note:** fact_inventory.sku_id is an integer surrogate key that maps to dim_product.product_sk — NOT to stg_prices.item_id directly. To join fact_inventory to pricing data, always route through dim_product: fact_inventory.sku_id → dim_product.product_sk → dim_product.item_id → stg_prices.item_id.

4. Data Dictionary

This section documents every column in every table in the DemandIQ NL data model. Columns are listed in ordinal position order matching information_schema.columns.

4.1 staging.stg_sales

Source: M5 sales_train_validation.csv — melted from wide to long format via chunked ETL. One row per item × store × date.

Column Name	Data Type	Nullable	Example Value	Description
item_id	VARCHAR	NO	FOODS_3_586	Original M5 product identifier. Maps to dim_product.item_id.
dept_id	VARCHAR	NO	FOODS_3	Department identifier from M5 dataset.
cat_id	VARCHAR	NO	FOODS	Category identifier. Mapped to Dutch nl_category in dim_product.
store_id	VARCHAR	NO	CA_3	Store identifier from M5. Maps to dim_store.store_id.
state_id	VARCHAR	NO	CA	State from M5 (US). Retained for traceability; not used in Dutch analytics.
date	DATE	NO	2020-01-01	Calendar date of the sales record. Maps to dim_date.full_date.
sales	FLOAT	YES	3.0	Units sold on this date. NULL allowed for dates with no sales data.

4.2 staging.stg_prices

Source: M5 sell_prices.csv. One row per item × store × week. Prices in euros (€) after Dutch market adaptation.

Column Name	Data Type	Nullable	Example Value	Description
store_id	VARCHAR	NO	CA_3	Store identifier. Join to dim_store.store_id.
item_id	VARCHAR	NO	FOODS_3_586	Product identifier. Join to dim_product.item_id.
wm_yr_wk	INTEGER	NO	11101	Walmart week number (YYYYY format). Join to dim_date for week lookup.
sell_price	TEXT	YES	2.0	Sell price in euros. Stored as TEXT in staging; cast to NUMERIC in warehouse views.

4.3 staging.stg_calendar

Source: M5 calendar.csv — adapted with Dutch public holiday flags. One row per calendar date.

Column Name	Data Type	Nullable	Example Value	Description
date	DATE	NO	2020-01-01	Calendar date.
wm_yr_wk	INTEGER	NO	11101	Walmart week number. Used to link prices to dates.
weekday	VARCHAR	NO	Wednesday	Day of week in English.
wday	INTEGER	NO	4	Day of week as integer (1=Sunday, 7=Saturday).
month	INTEGER	NO	1	Month number (1–12).
year	INTEGER	NO	2020	Calendar year.
event_name_1	VARCHAR	YES	Koningsdag	Dutch public holiday name, if applicable. NULL if no holiday.
event_type_1	VARCHAR	YES	National	Holiday type (National, Religious, Cultural).
event_name_2	VARCHAR	YES	NULL	Second event on same date (rare). NULL if none.
event_type_2	VARCHAR	YES	NULL	Type of second event. NULL if none.
snap_CA	INTEGER	NO	0	SNAP benefit flag (US data artefact). Retained; not used in Dutch analytics.
snap_TX	INTEGER	NO	0	SNAP TX flag. Retained; not used.
snap_WI	INTEGER	NO	0	SNAP WI flag. Retained; not used.
d	VARCHAR	NO	d_1	M5 day label (d_1 through d_1969). Used during staging ingestion only.

4.4 warehouse.dim_date

Conformed date dimension. One row per calendar date. All analytics date joins go through this table.

Column Name	Data Type	Nullable	Example Value	Description
date_sk	INTEGER	NO	20200101	Surrogate key. Format YYYYMMDD. Primary key.
full_date	DATE	NO	2020-01-01	Calendar date. Join target from fact tables.
day_of_week	VARCHAR	NO	Wednesday	Day name in English.
day_num	INTEGER	NO	4	Day of week (1=Sunday, 7=Saturday).
week_num	INTEGER	NO	1	ISO week number within year.
wm_yr_wk	INTEGER	NO	11101	Walmart week key. Bridge to stg_prices.
month_num	INTEGER	NO	1	Month number (1–12).
month_name	VARCHAR	NO	January	Month name in English.
quarter	INTEGER	NO	1	Quarter number (1–4).
year	INTEGER	NO	2020	Calendar year.
is_weekend	BOOLEAN	NO	FALSE	True if Saturday or Sunday.
is_holiday	BOOLEAN	NO	FALSE	True if Dutch public holiday.
holiday_name	VARCHAR	YES	Koningsdag	Dutch holiday name. NULL if not a holiday.
holiday_type	VARCHAR	YES	National	Holiday classification. NULL if not a holiday.

4.5 warehouse.dim_product

Product dimension. One row per SKU. Dutch category and department names applied.

Column Name	Data Type	Nullable	Example Value	Description
product_sk	SERIAL	NO	1	Integer surrogate key. Primary key.
item_id	VARCHAR	NO	FOODS_3_586	Original M5 item identifier. Bridge to stg_prices.item_id.
product_name	VARCHAR	NO	FOODS_3_586	Product display name (same as item_id in current model).
dept_id	VARCHAR	NO	FOODS_3	Original M5 department identifier.
dept_name	VARCHAR	NO	Voeding 3	Dutch department name (mapped from M5 dept_id).
cat_id	VARCHAR	NO	FOODS	Original M5 category identifier.

source_category	VARCHAR	NO	FOODS	Retained source category for audit purposes.
nl_category	VARCHAR	NO	Voeding	Dutch FMCG category: Voeding, Huishouden, Hobby & Vrije Tijd.
nl_department	VARCHAR	NO	Voeding 3	Dutch department name. Consistent with dept_name.
created_at	TIMESTAMP	NO	2024-01-01 00:00:00	Row creation timestamp.

4.6 warehouse.dim_store

Store dimension. One row per retail location. Dutch province and city mappings applied.

Column Name	Data Type	Nullable	Example Value	Description
store_sk	SERIAL	NO	1	Integer surrogate key. Primary key.
store_id	VARCHAR	NO	CA_3	Original M5 store identifier. Retained for data traceability.
store_name	VARCHAR	NO	Utrecht Central	Dutch store display name used in all analytics and dashboards.
city	VARCHAR	NO	Utrecht	Dutch city where the store is located.
province	VARCHAR	NO	Utrecht	Dutch province. Used for regional analytics in Power BI.
store_type	VARCHAR	NO	Supermarkt	Classification of retail format.
created_at	TIMESTAMP	NO	2024-01-01	Row creation timestamp.

4.7 warehouse.fact_inventory

Core output of the Phase 6 inventory engine. One row per SKU × Store combination. Contains safety stock, reorder point, and composite risk score. This table drives the Power BI Stockout Risk and Inventory Health pages.

Column Name	Data Type	Nullable	Example Value	Description
sku_id	INTEGER	NO	1	Foreign key → dim_product.product_sk.
store_sk	INTEGER	NO	1	Foreign key → dim_store.store_sk.
abc_class	VARCHAR	NO	A	ABC demand classification: A (top 80%), B (next 15%), C (remainder).

dept_name	VARCHAR	NO	Voeding 3	Department name denormalised from dim_product for query convenience.
avg_daily_demand	NUMERIC	NO	2.43	Average daily units sold across the full date range. Used as $\bar{d}$ in ROP formula.
std_daily_demand	NUMERIC	NO	1.87	Standard deviation of daily demand. Used as σ in safety stock formula.
cv	NUMERIC	NO	0.77	Coefficient of variation (std / avg). Demand variability indicator.
lead_time_days	INTEGER	NO	7	Assumed supplier lead time in days. Per-category default in current model.
service_level	NUMERIC	NO	0.95	Target service level. 0.95 = 95% (Z=1.65) across all SKUs.
z_score	NUMERIC	NO	1.65	Z-score corresponding to service_level. 1.65 for 95%.
safety_stock	NUMERIC	NO	5.47	Calculated safety stock in units: Z × σ × √L.
cycle_demand	NUMERIC	NO	17.01	Expected demand during lead time: $\bar{d}$ × L.
reorder_point	NUMERIC	NO	22.48	Reorder trigger in units: cycle_demand + safety_stock.
max_stock	NUMERIC	NO	39.49	Maximum recommended stock: reorder_point + safety_stock.
stockout_risk_sc	NUMERIC	NO	47.3	Composite risk score 0–100. See Section 6 for full logic.
risk_tier	VARCHAR	NO	MEDIUM	HIGH (≥50), MEDIUM (25–49), LOW (<25).
calculated_at	TIMESTAMP	NO	2024-01-01	Timestamp of engine run that produced this row.

5. PostgreSQL View Definitions

Five analytical views provide the data layer for Power BI. All views are loaded in Import mode via Npgsql v4.0.10. Views are defined in sql/07_powerbi_views.sql.

5.1 vw_executive_summary

Powers the Executive Summary dashboard page. Aggregates total revenue, units, SKU count, and monthly revenue trend.

Output Column	Type	Logic
total_revenue	NUMERIC	SUM(units_sold × sell_price::numeric) across full fact_sales
total_units	BIGINT	SUM(units_sold) across full fact_sales
total_skus	BIGINT	COUNT(DISTINCT product_sk) with any sales > 0
total_stores	BIGINT	COUNT(DISTINCT store_sk) with any sales > 0
month_year	TEXT	TO_CHAR(full_date, 'YYYY-MM') for trend grouping
monthly_revenue	NUMERIC	SUM revenue grouped by month_year
nl_category	VARCHAR	From dim_product — used for category revenue breakdown

5.2 vw_stockout_risk

Powers the Stockout Risk dashboard page. Joins fact_inventory to dim_product and dim_store to produce the full SKU × Store risk table.

Output Column	Type	Logic
sku_id	INTEGER	fact_inventory.sku_id
item_id	VARCHAR	dim_product.item_id — original M5 SKU code
product_name	VARCHAR	dim_product.product_name
nl_category	VARCHAR	dim_product.nl_category — Dutch category name
dept_name	VARCHAR	fact_inventory.dept_name
abc_class	VARCHAR	fact_inventory.abc_class
store_sk	INTEGER	fact_inventory.store_sk
store_name	VARCHAR	dim_store.store_name
province	VARCHAR	dim_store.province
risk_tier	VARCHAR	fact_inventory.risk_tier
stockout_risk_sc	NUMERIC	fact_inventory.stockout_risk_sc
safety_stock	NUMERIC	fact_inventory.safety_stock
reorder_point	NUMERIC	fact_inventory.reorder_point
avg_daily_demand	NUMERIC	fact_inventory.avg_daily_demand

5.3 vw_inventory_health

Powers the Inventory Health dashboard page. Aggregates inventory metrics by province and category.

Output Column	Type	Logic
province	VARCHAR	dim_store.province
nl_category	VARCHAR	dim_product.nl_category
abc_class	VARCHAR	fact_inventory.abc_class
avg_safety_stock	NUMERIC	AVG(safety_stock) per province × category
avg_reorder_point	NUMERIC	AVG(reorder_point) per province × category
avg_daily_demand	NUMERIC	AVG(avg_daily_demand) per province × category
high_risk_count	BIGINT	COUNT WHERE risk_tier = 'HIGH'
total_skus	BIGINT	COUNT of SKU × Store combinations in group
high_risk_pct	NUMERIC	high_risk_count / total_skus × 100

5.4 vw_regional_performance

Powers the Regional Performance dashboard page. Aggregates revenue and units by Dutch province and month.

Output Column	Type	Logic
province	VARCHAR	dim_store.province — Dutch province name
store_name	VARCHAR	dim_store.store_name
year	INTEGER	dim_date.year
month_num	INTEGER	dim_date.month_num
month_year	TEXT	TO_CHAR(full_date, 'YYYY-MM')
total_revenue	NUMERIC	SUM(units_sold × sell_price::numeric)
total_units	BIGINT	SUM(units_sold)
avg_daily_revenue	NUMERIC	total_revenue / COUNT(DISTINCT full_date)

5.5 vw_forecast_vs_actual

Status: This view is defined but not yet fully populated. It requires the XGBoost forecast results to be written back to PostgreSQL (a forecast_results table). The view definition exists in 07_powerbi_views.sql. Full activation is a medium-term recommendation.

Output Column	Type	Logic
item_id	VARCHAR	dim_product.item_id
full_date	DATE	dim_date.full_date
actual_sales	NUMERIC	fact_sales.units_sold
predicted_sales	NUMERIC	forecast_results.predicted_units (pending write-back)

`abs_error`	**NUMERIC**	ABS(actual_sales - predicted_sales)
`pct_error`	**NUMERIC**	abs_error / NULLIF(actual_sales, 0) × 100

6. Risk Score Architecture

6.1 Core Formulas

Safety Stock

```
Safety Stock = Z × σ × √L
```

Var	Value	Definition
Z	1.65	Z-score for 95% service level (industry standard for FMCG Class A SKUs).
σ	Per-SKU standard deviation	Calculated from STDDEV(units_sold) per item_id × store_id from stg_sales.
L	Per-category lead time (days)	Default lead times per category. Override when actual supplier data is available.

Reorder Point

```
Reorder Point = (d̄ × L) + Safety Stock
```

d̄ = AVG(units_sold) per item × store. L = lead time in days. Safety Stock = calculated above.

Composite Risk Score (0–100)

The composite risk score combines three sub-scores, each normalised to a 0–100 range and weighted:

Component	Weight	Formula	Business Meaning
Demand Variability	40%	CV = σ / d̄ × 100 (capped at 100)	High coefficient of variation = unpredictable demand = higher buffer needed.
Safety Stock Ratio	35%	(SS_calc / SS_actual) × 100 (capped at 100)	If actual SS is below calculated SS, ratio > 1 = increased risk.
Reorder Proximity	25%	(ROP / max_stock) × 100	How close current projected stock is to triggering a reorder.

```
Risk Score = (CV_score × 0.40) + (SS_ratio_score × 0.35) +
(proximity_score × 0.25)
```

6.2 Risk Tier Thresholds

Thresholds were calibrated against the actual score distribution after the initial run produced implausibly low results (threshold of ≥70 flagged only 6 combinations). Recalibration to ≥50 was validated by comparing HIGH-risk store concentration against store revenue rankings.

Tier	Threshold	Portfolio Count	Recommended Action

HIGH	Score ≥ 50	**3,071 (13.2%)**	Immediate demand planner review. Replenishment order likely required within 1–2 days.
MEDIUM	25 ≤ Score < 50	**Remainder**	Monitor weekly. Flag for next planning cycle. No urgent action required.
LOW	Score < 25	**—**	No action required. Review monthly as standard practice.

6.3 Lead Time Defaults

Until actual supplier lead time data is available, the following category defaults are used:

Dutch Category	Default Lead Time	Rationale
Voeding (Food)	**5 days**	High velocity; short replenishment cycles typical for Dutch food distributors.
Huishouden (Household)	**7 days**	Standard mid-range replenishment cycle.
Hobby & Vrije Tijd (Hobbies)	**10 days**	Slower velocity; longer acceptable lead time.
Persoonlijke Verzorging (Personal)	**7 days**	Standard cycle; similar to Household.

7. KPI Definitions

Every metric displayed on any Power BI dashboard page is defined below. KPIs are grouped by dashboard page. For each KPI: the business definition, the source view and column, the formula, the unit, and any edge cases or known limitations are documented.

7.1 Executive Summary

KPI Name	Definition	Source & Formula	Unit	Edge Cases
Total Revenue	Sum of all sales revenue across the full date range and all stores.	`vw_executive_summary SUM(units_sold × sell_price)`	€ (EUR)	NULL sell_price rows excluded. Prices cast from TEXT to NUMERIC.
Total Units	Sum of all units sold across full portfolio.	`vw_executive_summary SUM(units_sold)`	Units	Zero-sales dates included in fact_sales as 0, not NULL.
Total SKUs	Count of distinct products with any recorded sales.	`vw_executive_summary COUNT(DISTINCT product_sk)`	Count	SKUs with zero sales in all periods excluded.
Monthly Revenue	Revenue aggregated by calendar month for trend chart.	`vw_executive_summary GROUP BY TO_CHAR(date,YY-MM)`	€ (EUR)	Partial months at dataset boundaries included as-is.

7.2 Stockout Risk

KPI Name	Definition	Source & Formula	Unit	Edge Cases
High Risk SKUs	Count of SKU × Store combinations with risk score ≥ 50.	`vw_stockout_risk COUNT WHERE risk_tier=HIGH`	Count	Risk tier assigned at engine run time. Recalculate after demand changes.
% Portfolio at Risk	Percentage of all assessed SKU × Store combinations flagged HIGH.	`vw_stockout_risk high_count/total_count × 100`	%	Denominator = 23,311 (full fact_inventory population).
Avg Risk Score	Mean composite risk score across all SKU × Store combinations.	`vw_stockout_risk AVG(stockout_risk_sc)`	0–100	Unweighted mean. Skewed by large LOW-risk population.
Category Risk Rate	Percentage of HIGH-risk combinations per Dutch category.	`vw_stockout_risk high/total GROUP BY nl_category`	%	Voeding highest at 15.64% in current engine run.

7.3 Inventory Health

KPI Name	Definition	Source & Formula	Unit	Edge Cases
Avg Safety Stock	Mean safety stock units across all SKU × Store combinations.	`vw_inventory_health AVG(safety_stock)`	Units	Portfolio average: 5.47 units. Varies significantly by category.
Avg Reorder Point	Mean reorder point trigger across all SKU × Store combinations.	`vw_inventory_health AVG(reorder_point)`	Units	Portfolio average: 17.54 units.
Avg Daily Demand	Mean daily units sold per SKU × Store across the full period.	`vw_inventory_health AVG(avg_daily_demand)`	Units	Portfolio average: 2.43 units per SKU per day.
Excess Inventory	Value of inventory held above statistically calculated safety stock.	`05_business_value_calculation.sql €2.37M total`	€ (EUR)	Requires price join via dim_product. See Section 6.
Annual Holding Saving	Holding cost saving from replacing flat 14-day rule with model output.	`excess_value × 0.25 €593K total`	€ (EUR)	25% FMCG industry holding cost rate applied.

7.4 Regional Performance

KPI Name	Definition	Source & Formula	Unit	Edge Cases
Revenue by Province	Total revenue per Dutch province over the full period.	`vw_regional_performance SUM(total_revenue) by province`	€ (EUR)	Utrecht dominates at €21.8M (approx 80% of total).
Revenue Trend	Monthly revenue per province from 2020-2025.	`vw_regional_performance month_year + province group`	€ (EUR)	Useful for detecting diverging regional trajectories.
Units by Province	Total units sold per province.	`vw_regional_performance SUM(total_units) by province`	Units	Comparable to revenue for price-mix analysis.

7.5 Forecast vs Actual (pending)

KPI Name	Definition	Source & Formula	Unit	Edge Cases
Avg Actual Demand	Mean actual units sold per SKU per day in forecast period.	`vw_forecast_vs_actual AVG(actual_sales)`	Units	Current pilot: 31.91 units avg across 10 Class A SKUs.
Avg Predicted Demand	Mean XGBoost predicted units per SKU per day.	`vw_forecast_vs_actual AVG(predicted_sales)`	Units	Current pilot: 29.46 units. Requires write-back to activate.
Avg Abs Error	Mean absolute difference between actual and predicted.	`vw_forecast_vs_actual AVG(abs_error)`	Units	Current pilot: 7.44 units. MAPE = abs_error/actual x 100.
RMSE Improvement	Percentage reduction in RMSE of XGBoost vs Prophet baseline.	`Calculated offline in 04_forecasting.ipynb`	%	Current result: 22.7% improvement across 10 SKUs.

8. Power BI Page Wireframes

The following wireframes document the intended layout of each Power BI dashboard page. They serve as the design specification for rebuilding or modifying the dashboard. Each visual element is referenced by its type, data source, and position.

Layout notation: [VISUAL TYPE] Description. Position references are approximate (Top / Middle / Bottom, Left / Centre / Right).

8.1 Page: Home

Purpose: Navigation hub. Users arrive here first. Three navigation tiles route to each analytical page.

> **[HEADER BAR]** *Full-width green bar — Patel Mart B.V. logo + DemandIQ NL title | TOP*
> **[HERO TEXT]** *Large centred headline: "DemandIQ NL — Predictive Supply Chain Intelligence" | MIDDLE-LEFT*
> **[KPI SUMMARY ROW]** *4 KPI cards across: €27.30M Revenue | 657 At-Risk SKUs | €593K Saving | 22.7% Forecast Improvement | MIDDLE*
> **[NAV TILE 1]** *Button: "Stockout Risk →" — routes to Stockout Risk page | BOTTOM-LEFT*
> **[NAV TILE 2]** *Button: "Inventory Health →" — routes to Inventory Health page | BOTTOM-CENTRE*
> **[NAV TILE 3]** *Button: "Regional Performance →" — routes to Regional page | BOTTOM-RIGHT*
> **[FOOTER STRIP]** *Dark green strip — project credit + date | BOTTOM*

8.2 Page: Executive Summary

Purpose: Senior management overview. Revenue performance, SKU coverage, and demand trend. Source view: vw_executive_summary.

> **[NAV BAR]** *5-button navigation bar across top — links to all pages | TOP-FULL WIDTH*
> **[KPI CARD — Revenue]** *€27.30M | vw_executive_summary.total_revenue | TOP-LEFT*
> **[KPI CARD — Units]** *9M units | vw_executive_summary.total_units | TOP-CENTRE-LEFT*
> **[KPI CARD — SKUs]** *740 SKUs | vw_executive_summary.total_skus | TOP-CENTRE-RIGHT*
> **[KPI CARD — Stores]** *10 stores | vw_executive_summary.total_stores | TOP-RIGHT*
> **[SLICER — Category]** *Dropdown: All / Voeding / Huishouden / Hobby / Persoonlijke Verzorging | MIDDLE-LEFT*
> **[SLICER — Year]** *Year selector 2020–2025 | MIDDLE-LEFT*
> **[LINE CHART — Revenue Trend]** *X: month_year | Y: monthly_revenue | Colour by nl_category | MIDDLE-CENTRE*
> **[BAR CHART — Revenue by Category]** *X: nl_category | Y: total_revenue | Forest Green fill | BOTTOM-LEFT*
> **[TABLE — Top SKUs]** *Columns: item_id | nl_category | total_revenue | total_units | BOTTOM-RIGHT*

8.3 Page: Stockout Risk

Purpose: Primary tool for demand planners. Identifies HIGH-risk combinations by category, store, and SKU. Source view: vw_stockout_risk.

> **[NAV BAR]** *5-button navigation bar | TOP-FULL WIDTH*
> **[KPI CARD — High Risk Count]** *3,071 | COUNT WHERE risk_tier=HIGH | TOP-LEFT*
> **[KPI CARD — Portfolio % at Risk]** *13.17% | high/total × 100 | TOP-CENTRE-LEFT*
> **[KPI CARD — Avg Risk Score]** *32.71 | AVG(stockout_risk_sc) | TOP-CENTRE-RIGHT*
> **[KPI CARD — Avg Safety Stock]** *5.47 units | AVG(safety_stock) | TOP-RIGHT*
> **[SLICER — Risk Tier]** *Buttons: HIGH / MEDIUM / LOW | MIDDLE-LEFT*
> **[SLICER — Category]** *Dropdown: nl_category | MIDDLE-LEFT*
> **[SLICER — Province]** *Dropdown: province | MIDDLE-LEFT*
> **[BAR CHART — Risk by Category]** *X: nl_category | Y: COUNT(risk_tier=HIGH) | Sorted descending | MIDDLE-CENTRE*
> **[TABLE — SKU Detail]** *Columns: item_id | nl_category | store_name | province | risk_tier | stockout_risk_sc | safety_stock | reorder_point. Conditional formatting: HIGH=Red, MEDIUM=Amber, LOW=Green | BOTTOM-FULL WIDTH*

8.4 Page: Inventory Health

Purpose: Operations manager view. Safety stock levels, reorder points, and region × category risk matrix. Source view: vw_inventory_health.

> **[NAV BAR]** *5-button navigation bar | TOP-FULL WIDTH*
> **[KPI CARD — Avg Safety Stock]** *5.47 units | AVG(safety_stock) | TOP-LEFT*
> **[KPI CARD — Avg Reorder Point]** *17.54 units | AVG(reorder_point) | TOP-CENTRE-LEFT*
> **[KPI CARD — Avg Daily Demand]** *2.43 units | AVG(avg_daily_demand) | TOP-CENTRE-RIGHT*
> **[KPI CARD — Annual Saving]** *€593K | from 05_business_value_calculation.sql | TOP-RIGHT*
> **[SLICER — ABC Class]** *Buttons: A / B / C | MIDDLE-LEFT*
> **[SLICER — Province]** *Dropdown | MIDDLE-LEFT*
> **[MATRIX — Risk by Region × Category]** *Rows: province | Columns: nl_category | Values: high_risk_pct | Conditional colour scale | MIDDLE-CENTRE*
> **[BAR CHART — Safety Stock by Category]** *X: nl_category | Y: avg_safety_stock | BOTTOM-LEFT*
> **[SCATTER — Demand vs Risk Score]** *X: avg_daily_demand | Y: stockout_risk_sc | Colour: risk_tier | BOTTOM-RIGHT*

8.5 Page: Regional Performance

Purpose: Commercial team view. Revenue and units by Dutch province and trend over time. Source view: vw_regional_performance.

> **[NAV BAR]** *5-button navigation bar | TOP-FULL WIDTH*
> **[KPI CARD — Utrecht Revenue]** *€21.8M | SUM WHERE province=Utrecht | TOP-LEFT*
> **[KPI CARD — Total Revenue]** *€27.30M | SUM(total_revenue) all provinces | TOP-CENTRE*
> **[KPI CARD — Total Units]** *9M units | SUM(total_units) | TOP-RIGHT*
> **[SLICER — Year]** *Year 2020–2025 | MIDDLE-LEFT*

> **[SLICER — Province]** *Multi-select dropdown | MIDDLE-LEFT*
>
> **[MAP VISUAL (optional)]** *Netherlands choropleth — revenue shading by province | MIDDLE-CENTRE*
>
> **[BAR CHART — Revenue by Province]** *X: province | Y: total_revenue | Sorted descending | MIDDLE-RIGHT*
>
> **[LINE CHART — Revenue Trend by Province]** *X: month_year | Y: total_revenue | Colour: province | BOTTOM-FULL WIDTH*

9. Naming Conventions

Consistent naming across tables, columns, files, and views is essential for maintainability. The following conventions apply to all DemandIQ NL objects.

9.1 Database Object Naming

Object Type	Convention	Examples
Schema	`Lowercase, short`	staging, warehouse
Staging tables	`stg_ prefix + source name`	stg_sales, stg_prices, stg_calendar
Dimension tables	`dim_ prefix + entity name`	dim_date, dim_product, dim_store, dim_price
Fact tables	`fact_ prefix + subject`	fact_sales, fact_inventory
Views	`vw_ prefix + descriptive name`	vw_executive_summary, vw_stockout_risk
Columns — keys	`entity_sk for surrogate, entity_id for natural`	date_sk, product_sk, store_sk; item_id, store_id
Columns — flags	`is_ prefix + boolean name`	is_weekend, is_holiday
Columns — dates	`date noun or _at suffix for timestamps`	full_date, created_at, calculated_at
Columns — metrics	`Descriptive noun or avg_/sum_ prefix`	avg_daily_demand, safety_stock, total_revenue

9.2 File Naming (Repository)

File Pattern	Examples
NN_descriptive_name.sql — SQL scripts numbered in execution order	01_star_schema.sql, 05_business_value_calculation.sql, 07_powerbi_views.sql
NN_descriptive_name.ipynb — Notebooks numbered in execution order	01_data_loading_staging.ipynb, 04_forecasting.ipynb, 05_inventory_engine.ipynb
PascalCase.pbix — Power BI files	DemandIQ_NL.pbix
UPPER_SNAKE_CASE.md — Documentation files	README.md, ARCHITECTURE.md

9.3 Power BI Naming

Object	Convention	Examples
Page names	Title Case, descriptive	Executive Summary, Stockout Risk, Inventory Health

Measure names	Title Case with unit if ambiguous	Total Revenue (€), Avg Safety Stock (units), High Risk %
Visual titles	Sentence case, specific	Revenue by Dutch province, Risk score by category
Slicer labels	Noun only — field name	Category, Province, Risk Tier, Year
Table columns	Readable label (not raw column name)	SKU Code (not item_id), Risk Score (not stockout_risk_sc)

10. Glossary

Definitions of all domain, technical, and Dutch-specific terms used throughout the DemandIQ NL system and documentation.

Term	Definition
ABC Classification	Demand-based product segmentation: Class A = top 80% of revenue (997 SKUs), Class B = next 15%, Class C = remainder. Determines service level priorities.
Avg Daily Demand ($\bar{d}$)	Arithmetic mean of daily units sold for a specific SKU × Store combination over the full date range. Core input to reorder point and safety stock calculations.
Bevrijdingsdag	Dutch Liberation Day, 5 May. National public holiday. Applied as demand suppression flag in dim_date.
Coefficient of Variation (CV)	Standard deviation divided by mean. Measures relative demand variability. High CV = more unpredictable demand = larger safety stock required.
Cycle Demand	Expected units sold during the replenishment lead time: $\bar{d} \times L$. Represents the inventory consumed while waiting for a new order to arrive.
DemandIQ NL	The DemandIQ NL predictive supply chain analytics system. Seven-phase pipeline from raw data to Power BI executive dashboard, built at Patel Mart B.V.
EDA (Exploratory Data Analysis)	Phase 4 of the DemandIQ NL pipeline. Statistical and visual analysis of the demand dataset to identify patterns, anomalies, and business-relevant segments before modelling.
FMCG	Fast-Moving Consumer Goods. Products consumed regularly and purchased frequently — food, household products, personal care. Characteristic of high-volume, low-margin distribution.
Holding Cost Rate	The annual cost of holding one euro of inventory, expressed as a percentage of inventory value. Industry standard for FMCG: 25%. Covers storage, capital cost, spoilage, and insurance.
Huishouden	Dutch for Household. One of the four Patel Mart product categories. Maps from M5 HOUSEHOLD category.
Hobby & Vrije Tijd	Dutch for Hobbies & Leisure. One of the four Patel Mart product categories. Maps from M5 HOBBIES category.
Kerst	Dutch Christmas (25–26 December). Applied as demand suppression flag in dim_date.
Koningsdag	Dutch King's Day, 27 April. Applied as demand shift flag in dim_date — consumer purchasing moves to outdoor/leisure categories.
Lead Time (L)	The number of days between placing a replenishment order and receiving the goods. Current model uses category-level defaults. See Section 6.3.
M5 Dataset	The M5 Forecasting Competition dataset from Kaggle. Original source data underlying DemandIQ NL. US Walmart retail sales transformed to Dutch FMCG distributor context.
MAPE	Mean Absolute Percentage Error. Forecast accuracy metric: mean of ABS(actual - predicted) / actual × 100. Lower is better. XGBoost achieved 21.3% average improvement over Prophet.
Max Stock	Maximum recommended inventory level per SKU × Store: Reorder Point + Safety Stock. Upper bound for optimal inventory holding.
Persoonlijke Verzorging	Dutch for Personal Care. Fourth Patel Mart product category.

Prophet	Open-source time series forecasting tool developed by Meta. Uses additive decomposition (trend + seasonality + holidays). Benchmarked against XGBoost in Phase 5. Lost on all 10 evaluated SKUs.
Reorder Point (ROP)	The inventory level at which a replenishment order must be placed: ($\bar{d}$ × L) + Safety Stock. Falling below the ROP without a pending order is an immediate stockout risk signal.
RMSE	Root Mean Square Error. Forecast accuracy metric penalising large errors more than MAPE. XGBoost achieved 22.7% average improvement over Prophet.
Safety Stock (SS)	Buffer inventory held above the expected demand during lead time to protect against demand variability. Formula: Z × σ × √L. Replaces the manual 14-day flat rule.
Service Level	The probability of not experiencing a stockout during a replenishment cycle. Set at 95% (Z = 1.65) for all Class A SKUs in the current model.
Sinterklaas	Dutch Sinterklaas celebration, 5 December. Applied as demand event flag — uplift in Hobby & Vrije Tijd, followed by demand depression.
SKU	Stock Keeping Unit. A unique product identifier at the item level. In DemandIQ NL: 3,049 distinct SKUs across all categories.
SKU × Store	The fundamental grain of the fact_inventory table. One row per unique combination of SKU and store location: 23,311 combinations assessed.
Star Schema	Dimensional data model with one central fact table joined to multiple dimension tables. Optimises for analytical query performance and BI tool compatibility.
Staging Layer	The first database layer in DemandIQ NL. Raw CSV data ingested with minimal transformation. Preserves source data for audit and reprocessing.
Stockout	An inventory event where a SKU reaches zero units. Results in lost sales, emergency reorder cost, and potential customer attrition. Primary risk event DemandIQ NL is designed to prevent.
Surrogate Key (_sk)	System-generated integer primary key assigned by the warehouse layer, independent of source system identifiers. Used for all fact-to-dimension joins.
Voeding	Dutch for Food. The highest-revenue and highest-risk product category at Patel Mart. Maps from M5 FOODS category.
Warehouse Layer	The second database layer. Star schema transformation of staging data: 37.2M fact rows, 4 dimension tables, optimised for analytical query and BI connectivity.
XGBoost	eXtreme Gradient Boosting. Gradient boosted tree algorithm used for demand forecasting in Phase 5. Outperformed Prophet on all 10 evaluated Class A SKUs.
Z-Score	Statistical measure corresponding to a given service level in a normal distribution. Z = 1.65 for 95% service level. Used in safety stock formula.

— End of Blueprint Document —

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